



MIDTERM EXAMINATION IN ISMAT3000L– QUANTITATIVE METHODS
2nd Semester, AY 2025-2026

Format:

- 1. Form a group (with max of 9 members).
- 2. Check the attached file here.
- 3. Put your answer in each number in MS Word but save it in PDF format.
- 4. Leader will submit it in our Google Classroom.
- 5. Deadline: March 28, 2026

Test I. Using Excel or Jamovi do the following:

OUTPUT 1: Trade winds are one of the beautiful features of island life in Hawaii. The following data represent total air movement in miles per day over a weather station in Hawaii as determined by a continuous anemometer recorder. The period of observation is January 1 to February 15, 1971.

26 14 18 14 113 50 13 22 27 57 28 50 72 52 105 138 16 33 18 16 32 26 11
17 14 57 100 35 20 21 34 18 13 18 28 21 13 25 19 11 19 22 19 15 20 16

Source: United States Department of Commerce, National Oceanic and Atmospheric Administration, Environmental Data Service. *Climatological Data, Annual Summary, Hawaii*, Vol. 67, No. 13. Asheville: National Climatic Center, 1971pp. 11, 24.

- (a) Use the computer to find the sample mean, median, and (if it exists) mode. Also, find the range, sample variance, and sample standard deviation.
- (b) Use the five-number summary provided by the computer to make a box-and-whisker plot of total air movement over the weather station.
- (c) Four data values are exceptionally high: 113, 105, 138, and 100. The strong winds of January 5 (113 reading) brought in a cold front that dropped snow on Haleakala National Park (at the 8000 ft elevation). The residents were so excited that they drove up to see the snow and caused such a massive traffic jam that the Park Service had to close the road. The winds of January 15, 16, and 28 (readings 105, 138, and 100) accompanied a storm with funnel clouds that did much damage. Eliminate these values (i.e., 100, 105, 113, and 138) from the data bank and redo parts (a) and (b). Compare your results with those previously obtained. Which average is most affected? What happens to the standard deviation? How do the two box-and-whisker plots compare?

OUTPUT 2: In West Texas, water is extremely important. The following data represent pH levels in ground water for a random sample of 102 West Texas wells. A pH less than 7 is acidic and a pH above 7 is alkaline. Scanning the data, you can see that water in this region tends to be hard (alkaline). Too high a pH means the water is unusable or needs expensive treatment to make it useable (Reference: C. E. Nichols and V. E. Kane, Union Carbide Technical Report K/UR-1).

x: pH of Ground Water in 102 West Texas Wells

7.0 7.0 7.0 7.0 7.0 7.0 7.0 7.0 7.1 7.1 7.1 7.1 7.1 7.1 7.1 7.1 7.1 7.1 7.2 7.2 7.2 7.2 7.2
7.2 7.2 7.2 7.2 7.2 7.3 7.3 7.3 7.3 7.3 7.3 7.3 7.3 7.3 7.3 7.3 7.4 7.4 7.4 7.4 7.4 7.4 7.4
7.4 7.5 7.5 7.5 7.5 7.5 7.5 7.5 7.6 7.6 7.6 7.6 7.6 7.6 7.6 7.6 7.6 7.7 7.7 7.7 7.7 7.7 7.7
7.8 7.8 7.8 7.8 7.8 7.9 7.9 7.9 7.9 7.9 8.0 8.1 8.1 8.1 8.1 8.1 8.1 8.1 8.2 8.2 8.2 8.2 8.2
8.2 8.2 8.4 8.5 8.6 8.7 8.8 8.8

- (a) Write a brief description in which you outline how you would obtain a random sample of 102 West Texas water wells. Explain how random numbers would be used in the selection process.
- (b) Is the given data nominal, ordinal, interval, or ratio? Explain.



- (c) Make a stem-and-leaf display.
- (d) Make a frequency histogram using any software.
- (e) Compute the range, mean, median, and mode for the given data using any software.
- (f) Compute the sample variance, and sample standard deviation for the given data using any software. Is the sample standard deviation small relative to the mean pH?
- (g) Make a box- -and-whisker plot using Jamovi. Find the interquartile range, lower fence, and upper fence. Determine if there is an outlier/s.

OUTPUT 3. Coal is carried from a mine in West Virginia to a power plant in New York in hopper cars on a long train. The automatic hopper car loader is set to put 75 tons of coal into each car. The actual weights of coal loaded into each car are *normally distributed*, with mean $\mu = 75$ tons and standard deviation $\sigma = 0.8$ ton.

- (a) What is the probability that one car chosen at random will have less than 74.5 tons of coal?
- (b) What is the probability that 20 cars chosen at random will have a mean load weight of less than 74.5 tons of coal?

OUTPUT 4. At Burnt Mesa Pueblo, the method of tree ring dating gave the following years A.D. for an archaeological excavation site (*Bandelier Archaeological Excavation Project: Summer 1990 Excavations at Burnt Mesa Pueblo*, edited by Kohler, Washington State University):

1189	1271	1267	1272	1268
1316	1275	1317	1275	

- (a) Use any software with mean and standard deviation keys to verify that the sample mean year is $\bar{x} = 1272$, with sample standard deviation $s = 37$ years.
- (b) Find a 95% confidence interval for the mean of all tree ring dates from this archaeological site.





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